

Dated:

Q2.

1. Generating Initial Population

Randomly assigning f_0, f_1, f_2 (Facility 1, 2, 3) to each task
each chromosome will have 7 genes - 7 tasks

Chromosome	Task Assignment
C1	[0, 0, 1, 2, 1, 2, 2]
C2	[1, 1, 1, 2, 0, 2, 0]
C3	[2, 2, 0, 0, 1, 1, 2]
C4	[0, 1, 0, 1, 2, 0, 2]
C5	[2, 0, 1, 0, 2, 1, 1]
C6	[1, 2, 2, 0, 0, 1, 2]

2. Evaluate fitness for each chromosome

Fitness: Time (Hr) $\times$ Cost

Penalty = 1000 (if exceeds)

C1 [0, 0, 1, 2, 1, 2, 2]

Task	Facility	Cost/Hr Time	Time Cost/Max	Total cost
T1	0 0	10	5	50
T2	0 0	15	8	120
T3	1 1	9	4	36
T4	2 2	13	7	91
T5	1 1	13	6	78
T6	2 2	10	3	30
T7	2 2	13	9	117

Facility Usage:

$$F_0: 5 + 8 = 13 \text{ hrs} < 24 \quad \text{valid}$$

$$F_1: 4 + 6 = 10 \text{ hrs} < 30 \quad \text{valid}$$

$$F_2: 7 + 3 + 9 = 19 \text{ hrs} < 28 \quad \text{valid}$$

$$\text{Total cost} = 50 + 120 + 36 + 91 + 78 + 30 + 117 = 522$$

// same for C2, C3, C4, C5, C6

Dated:

Chromosome	Assignment	Fitness
C1	[0,0,1,2,1,2,2]	522
C2	[1,1,1,2,0,2,0]	540
C3	[2,2,0,0,1,1,2]	585
C4	[0,1,0,1,2,0,2]	510
C5	[2,0,1,0,2,1,1]	498
C6	[1,2,2,0,0,1,2]	505
Sum		3160

3. Selection (Roulette Wheel)

Chromosome	Fitness	$1/\text{Fitness}$	Selection Probability
C1	522	0.0019157	16.76%
C2	540	0.0018519	16.21%
C3	585	0.0017094	14.95%
C4	510	0.0019608	17.16%
C5	498	0.0020080	17.57%
C6	505	0.0019802	17.33%
Sum		0.01142598	100%

$\frac{1}{\text{Fitness}}$ = Invert Fitness = because we're minimizing cost to give higher selection chance to lower cost chromosomes

$$\text{Selection Probability: } P_i = \frac{1/F_i}{\text{Total Inverted Fitness}}$$

// Spin Roulette wheel 6 times to get random no. from 0-1 and determine which chromosome to pick based on cumulative probability

Chromosome	Cumulative Probability
C1	0.1676
C2	0.3297
C3	0.4792
C4	0.6508
C5	0.8265
C6	1

Dated:

0.75 $\rightarrow$ C5

0.62 $\rightarrow$ C4

0.91 $\rightarrow$ C6

0.7 $\rightarrow$ C5

0.1 $\rightarrow$ C1

0.55 $\rightarrow$ C4

$\uparrow$

Random
values

Final Selection

4. ^{Crossover} Mutation (One-point & 80%.)

Pair 1 : C5 & C4

Pair 2 : C6 & C5

Pair 3 : C1 & C4

$\rightarrow$ Generate random no. $0 \leq x \leq 1$
if $x \leq 0.8$; perform crossover
if $x > 0.8$; skip crossover

Pair 1

C5 : [2, 0, 1, 0, 2, 1, 1]

C4 : [0, 1, 0, 1, 2, 0, 2]

$x = 0.52$

apply crossover

Child 1 = [2, 0, 1, 1, 2, 0, 2]

Child 2 = [0, 1, 0, 0, 2, 1, 1]

Pair 2

C6 : [1, 0, 2, 1, 2, 0, 1]

C5 : [2, 0, 1, 0, 2, 1, 1]

$x = 0.78$

apply crossover

Child 3 = [1, 0, 2, 0, 2, 1, 1]

Child 4 = [2, 0, 1, 1, 2, 0, 1]

Pair 3

C1 : [1, 1, 0, 1, 0, 2, 0]

C4 : [0, 1, 0, 1, 2, 0, 2]

$x = 0.6$

apply crossover

Child 5 = [1, 1, 0, 1, 2, 0, 2]

Child 6 = [0, 1, 0, 1, 0, 2, 0]

Dated:

5. Mutation (20% & swap mutation)

$x \leq 0.2$ mutation
 $x > 0.2$ no mutation

Child	x	Mutation?	
C1	0.13	✓	
C2	0.41	X	unchanged
C3	0.09	✓	
C4	0.81	X	unchanged
C5	0.19	✓	
C6	0.76	X	unchanged

C1: [2, 0, 1, 1, 2, 0, 2] swap index 1 & 4
 [2, 2, 1, 1, 0, 0, 2]

C3: [1, 0, 2, 0, 2, 1, 1] swap index 2 & 6
 [1, 0, 1, 0, 2, 1, 2]

C5: [1, 1, 0, 1, 2, 0, 2] swap index 0 & 5
 [0, 1, 0, 1, 2, 1, 2]

6. Evaluating Fitness of New Generation

C1: [2, 2, 1, 1, 0, 0, 2]

Task	Facility	Time	Cost/Hr	Total Cost
T1	2	5	9	45
T2	2	8	16	128
T3	1	4	9	36
T4	1	7	10	70
T5	0	6	14	84
T6	0	3	9	27
T7	2	9	13	117

for

Dated:

Facility Usage:

$$F0 : 6 + 3 = 9 < 24$$

valid

$$F1 : 4 + 7 = 11 < 30$$

valid

$$F2 : 5 + 9 = 22 < 28$$

valid

Total cost = 507

// same for C2, C3, C4, C5, C6

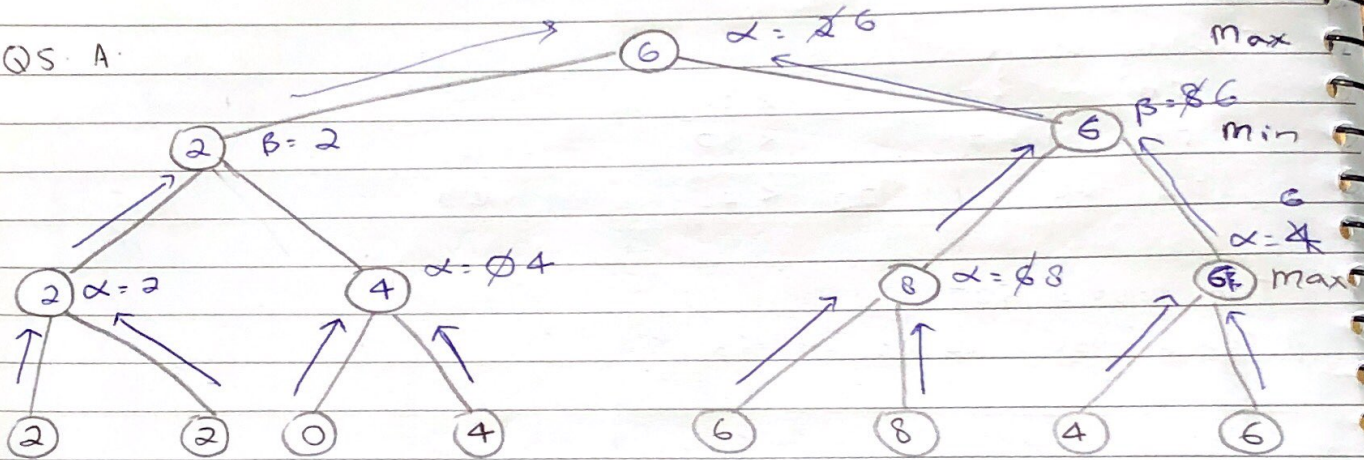
Child	Chromosome	Fitness
C1	[2, 2, 1, 1, 0, 0, 2]	507
C2	[0, 1, 0, 0, 2, 1, 1]	508
C3	[1, 0, 2, 2, 1, 1]	505
C4	[2, 0, 1, 1, 2, 0, 1]	511
C5	[1, 1, 0, 1, 2, 0, 2]	497
C6	[0, 1, 0, 1, 0, 2, 0]	520
Sum		3048

3048 < 3160 optimized

Best solution = C5 : 497

Dated:

Q5. A.



- ~~1/2~~ Max $\alpha = 2$; explored 2 & 2

- Min $\beta = 2$

- Max $\alpha = 0$; explored 0
explored 4 so $\alpha = 4$

- Max $\alpha = 2$

- Max $\alpha = 6$; explored 6
explored 8 so $\alpha = 8$

- Min $\beta = 8$

- Max $\alpha = 4$; explored 4
explored 6 so $\alpha = 6$

- so Min $\beta = 6$

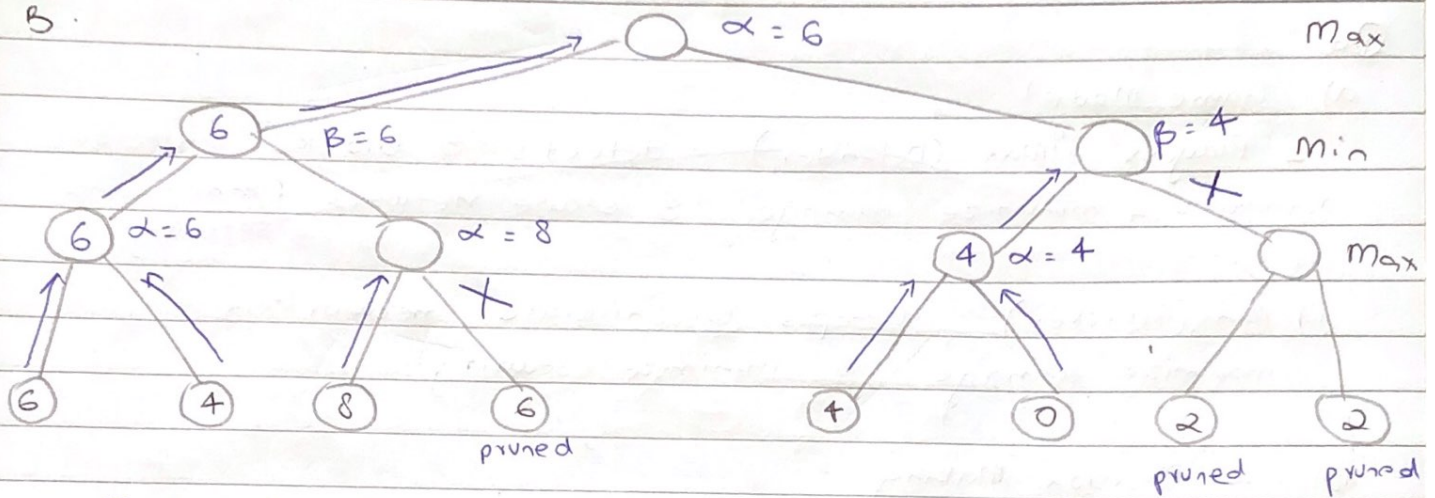
$\therefore$ No parts were cut off (no pruning)

Winning path $\alpha = 6 \rightarrow \beta = 6, \rightarrow \alpha = 6$

- so Max $\alpha = 6$

Dated:

B.



- Max $\alpha = 6$; explored 6 & 4

- Min $\beta = 6$

- Max $\alpha = 8$ but $\beta = 6$ so prune 6

- Max $\alpha = 6$

- Max $\alpha = 4$; explored 4 & 0

- Min $\beta = 4$ but $\alpha = 6$ so prune 2 & 2

∴ Values that were cut off : 6, 2, 2

Winning Path : $\alpha = 6 \rightarrow \beta = 6 \rightarrow \alpha = 6$

Dated:

Q6.

a) Game Model

i) Max (Defender) : detects & blocks malicious activities, minimize damage & secure network (maximizing security)

ii) Min (Attacker) : attempts to infiltrate network, maximize damage & minimize security

2. Decision Making

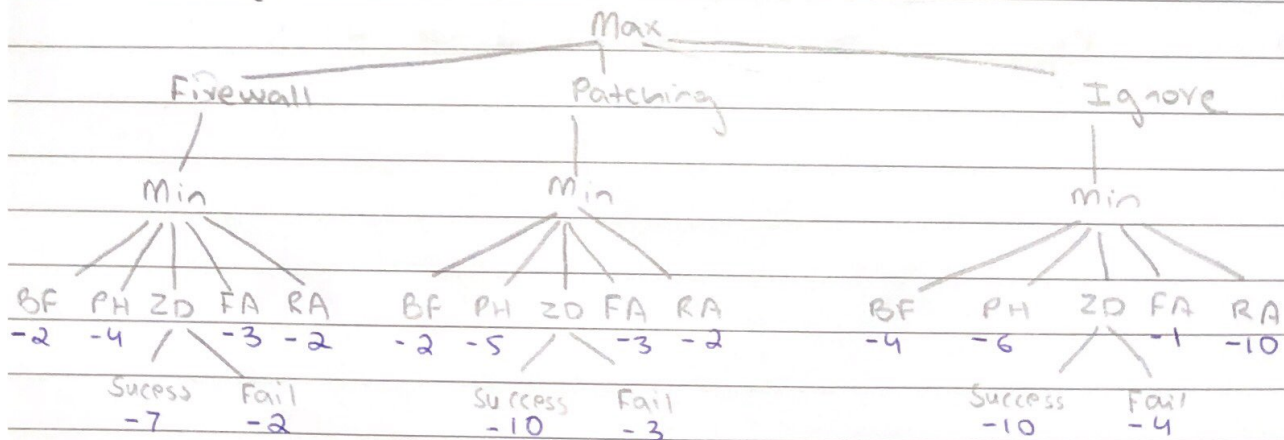
- Max : observes behaviour / identifies attack & chooses option where Max has high success
- Min : identifies weak points, chooses options where Max has low success - may use fake attacks to manipulate defender's resource allocation

3. Stochastic Elements

Probabilistic Outcomes - Zero Day Exploits - chance nodes (unpredictable)

- Max can't guarantee safe outcome - so expectation based strategy (instead of minimax) where Max weighs expected damage

b) Constructing Game Tree



Dated:

Utility values for attack

BF -2

PH -5

ZD -10 (success) -3 (fail)

FA -3

RA -10 (if ignored) -3 (detected)

c) 1. MiniMax

Assuming ZD = -4.5 (in firewall scenario)

ZD = -6.5 (in Patch system)

ZD = -7 (in Ignore)

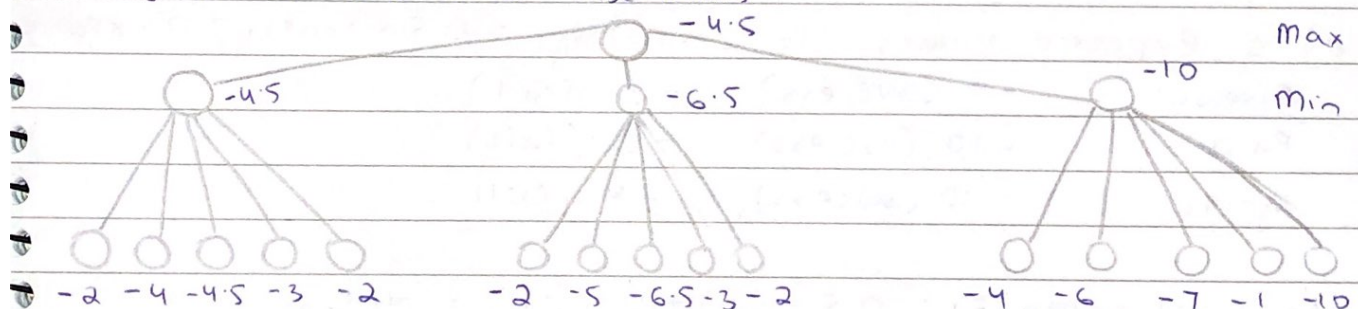
↳ Calculated Expected utility

if min chooses -4.5, Max chooses Firewall

if min chooses -6.5, Max chooses Patch System

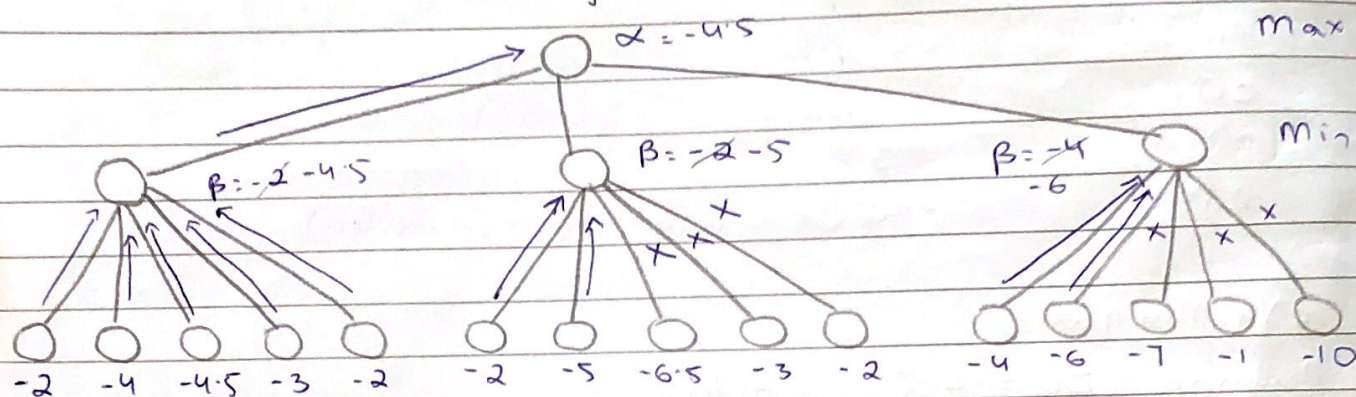
if min chooses -10, Max ignores

so best move for max is to deploy firewall as it is the highest of the worst case values



Dated:

2. Alpha beta pruning



- $\beta = -2$ ~~-4~~ -4.5 explored $-2, -4, -4.5, -3, -2$

• so $\alpha = -4.5$

- $\beta = -2, -5$ but $\alpha = -4.5$ so
prune $-6.5, -3, -2$

- $\beta = -4, -6$ but $\alpha = -4.5$ so
prune $-7, -1, -10$

d) 1. Expected values for Zero Day = $0.5 \times \text{Success} + 0.5 \times \text{Failure}$

Firewall : -7 (success) -2 (fail)

Patch : -10 (success) -3 (fail)

Ignore : -10 (success) -4 (fail)

Expected value (F) : $0.5 \times (-7) + 0.5 \times (-2) = -4.5$

Expected value (P) : $0.5 \times (-10) + 0.5 \times (-3) = -6.5$

Expected value (I) : $0.5 \times (-10) + 0.5 \times (-4) = -7$

2. In Firewall : $EV = \frac{-2-4-4.5-3-2}{5} = -3.1$

Patch : $EV = -3.7$

; In Ignore : $EV = -5.6$

Max should still choose Deploy Firewall

Dated:

Q4.

a) Leftmost

X	X	
X		
0	0	

i)

X	X	0
X		
0	0	

$$R_0: X X 0 \rightarrow v=0, C_0: X X 0 \rightarrow v=0$$

$$R_1: X \rightarrow v=10, C_1: X 0 \rightarrow v=0$$

$$R_2: 0 0 \rightarrow v=-100, C_2: 0 \rightarrow v=-10$$

$$D_1: X \rightarrow v=10, D_2: 0 0 \rightarrow v=-100$$

$$V = 10 - 100 - 10 + 10 - 100 = -190$$

ii)

X	X	
X	0	
0	0	

$$R_0: X X \rightarrow v=100, C_0: X X 0 \rightarrow v=0$$

$$R_1: X 0 \rightarrow v=0, C_1: X 0 0 \rightarrow v=0$$

$$R_2: 0 0 \rightarrow v=-100, C_2: \emptyset \rightarrow v=0$$

$$D_1: X 0 \rightarrow v=0, D_2: 0 0 \rightarrow v=-100$$

$$V = 100 - 100 - 100 = -100$$

iii)

X	X	
X		0
0	0	

$$R_0: X X \rightarrow v=100, C_0: X X 0 \rightarrow v=0$$

$$R_1: X 0 \rightarrow v=0, C_1: X 0 \rightarrow v=0$$

$$R_2: 0 0 \rightarrow v=-100, C_2: 0 \rightarrow v=-10$$

$$D_1: X \rightarrow v=10, D_2: 0 \rightarrow v=-10$$

$$V = 100 - 100 - 10 + 10 - 10 = -10$$

Dated:

iv)

X	X	
X		
0	0	0

R0 : XX $\rightarrow$ V=100 , C0 : XX 0 $\rightarrow$ V=0

R1 : X $\rightarrow$ V=10 , C1 : X 0 $\rightarrow$ V=0

R2 : 000 $\rightarrow$ V=-1000, C2 : 0 $\rightarrow$ V=-10

D1 : X 0 $\rightarrow$ V=0 , D2 : 0 $\rightarrow$ V=-10

~~V = 100 + 10 - 1000~~ R2 has 3 0s so V=-1000 (Worst)

b) Rightmost

X		
X		
0	0	X

i)

X	0	
X		
0	0	X

R0 : X 0 $\rightarrow$ V=0 , C0 : X X 0 $\rightarrow$ V=0

R1 : X $\rightarrow$ V=10 , C1 : 0 0 $\rightarrow$ V=-100

R2 : 0 0 X $\rightarrow$ V=0 , C2 : X $\rightarrow$ V=10

D1 : X X $\rightarrow$ V=100 , D2 : 0 $\rightarrow$ V=-10

$$V = 10 - 100 + 10 + 100 - 10 = 10$$

ii)

X		0
X		
0	0	X

R0 : X 0 $\rightarrow$ V=0 , C0 : X X 0 $\rightarrow$ V=0

R1 : X $\rightarrow$ V=10 , C1 : 0 $\rightarrow$ V=-10

R2 : 0 0 X $\rightarrow$ V=0 , C2 : 0 X $\rightarrow$ V=0

D1 : X X $\rightarrow$ V=100 , D2 : 0 0 $\rightarrow$ V=-100

$$V = 10 + 100 - 10 - 100 = 0$$

iii)

X		
X	0	
0	0	X

R0 : X $\rightarrow$ V=10 , C0 : X X 0 $\rightarrow$ V=0

R1 : X 0 $\rightarrow$ V=0 , C1 : 0 0 $\rightarrow$ V=-100

R2 : 0 0 X $\rightarrow$ V=0 , C2 : ~~0 0~~ X $\rightarrow$ V=10

D1 : X 0 X $\rightarrow$ V=0 , D2 : 0 0 $\rightarrow$ V=-100

$$V = 10 - 100 + 10 - 100 = -180$$

Dated:

iv)

X		
X		O
O	O	X

$R_0 : X \rightarrow V = 10$, $C_0 : X \times O \rightarrow V = 0$

$R_1 : X \ O \rightarrow V = 0$, $C_1 : \quad O \rightarrow V = -10$

$R_2 : O \ O \ X \rightarrow V = 0$, $C_2 : \quad O \ X \rightarrow V = 0$

$D_1 : X \ X \rightarrow V = +100$, $D_2 : \quad O \rightarrow V = -10$

$$V = 10 - 10 + 100 - 10 = 90$$

In leftmost ; best case = -10

worst case = -1000

In rightmost ; best case = 90

worst case = -180

~~in~~ X should choose rightmost move at (2,2)

X		
X		
O	O	X

; leads to better worst case

$$-180 > -1000$$